Problem Statement 1: **Federated Multimodal Edge–Cloud Learning for Low-Latency, Privacy-Preserving IoT**

IoT systems increasingly combine diverse sensors (e.g. cameras, microphones, environmental probes) to improve situational awareness. However, sending all raw multimodal data to the cloud incurs high latency and communication costs. On-device inference can "reduce the volume of transmitted data and address privacy concerns by keeping sensitive processing local". We propose a novel hierarchical federated learning framework that jointly trains AI models across devices, edge gateways, and the cloud to satisfy strict QoS (latency, throughput, reliability) while enforcing privacy. The system will use two-tier aggregation (device → edge → cloud) to cut communication load and will inject calibrated differential-privacy noise in local updates to protect data. This approach is applicable to many domains: e.g. IIoT can fuse camera and vibration data, smart cities can combine traffic video and pollution readings, and healthcare can merge wearable and imaging data, all requiring real-time inference with strong privacy.

Objectives:

- **Hierarchical FL Design:** Develop a two-tier federated learning algorithm for multimodal data (e.g. combining image, audio, and sensor features) and quantify communication reduction (target ≥50% reduction in uplink traffic vs. centralized training) and latency improvements through simulation.
- **Privacy Mechanisms:** Integrate differential privacy or secure aggregation in local updates, ensuring a strict privacy budget (e.g. $\varepsilon \leq 1$) while limiting model accuracy loss to ≤2%.
- **QoS Validation:** Prototype the system (using tools like EdgeCloudSim/NS-3) to achieve inference latency below a target threshold (e.g. <100 ms) and high reliability (>99% successful model updates) under realistic network conditions.
- **Energy Efficiency:** Measure end-to-end energy consumption and demonstrate a ≥20% energy saving (per training round) by balancing local computation versus communication.

Problem 1 — Federated Multimodal Edge–Cloud Learning

Brief problem definition

Multimodal IoT devices generate large, heterogeneous data that overloads networks and risks privacy when centralized.
Transmitting raw multimodal streams to cloud increases latency and energy use, harming real-time decisions.
On-device models reduce data transfer but struggle to remain accurate and coordinated across devices.
We need a hierarchical, privacy-preserving learning approach that balances local computation, communication, latency and energy across edge–cloud layers.

Problem statement

Hierarchical federated multimodal learning for low-latency, privacy-preserving edge–cloud IoT.

Objectives

1. Reduce uplink data by at least fifty percent via two-tier aggregation.
2. Limit model accuracy loss to ≤2% while preserving privacy budget.
3. Achieve inference latency below 100 milliseconds across heterogeneous networks.